

How neighbouring field lines turn: direction and azimuthal mean

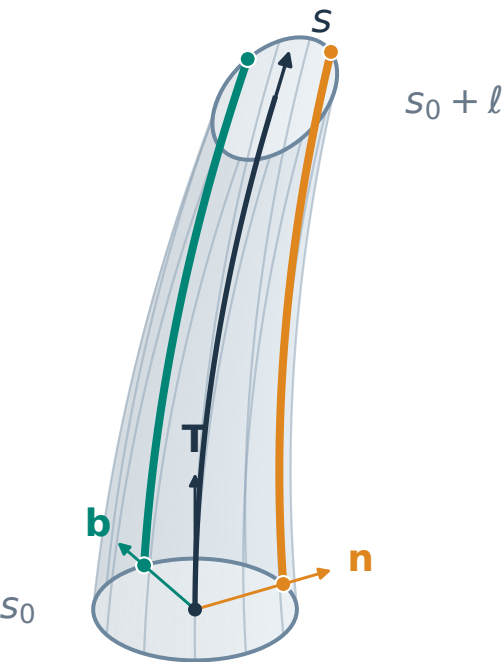
Rotation per unit arc length, measured in a transported normal plane

$$\mathbf{T} = \mathbf{B}/B, \quad \mathbf{b} = \mathbf{T} \times \mathbf{n}$$

$$\Omega(\phi) = \mathbf{T} \cdot [\mathbf{e} \times ((\mathbf{e} \cdot \nabla)\mathbf{T})] = \frac{\alpha}{2} + \frac{\beta_g}{2} \cos 2\phi - \frac{\delta_g}{2} \sin 2\phi$$

1 NEIGHBOURING FIELD LINES

$$M = \begin{bmatrix} a & q \\ p & d \end{bmatrix}$$

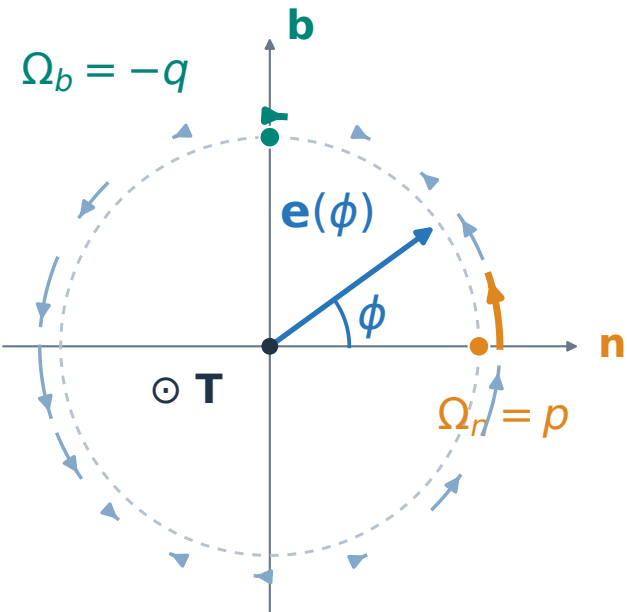


Dark: reference Colour: neighbours

$\ell = 0.9L_0$: arc length between sections along the reference line

2 DIRECTIONAL ANGULAR RATES

$$\mathbf{e}(\phi) = \cos \phi \mathbf{n} + \sin \phi \mathbf{b}$$

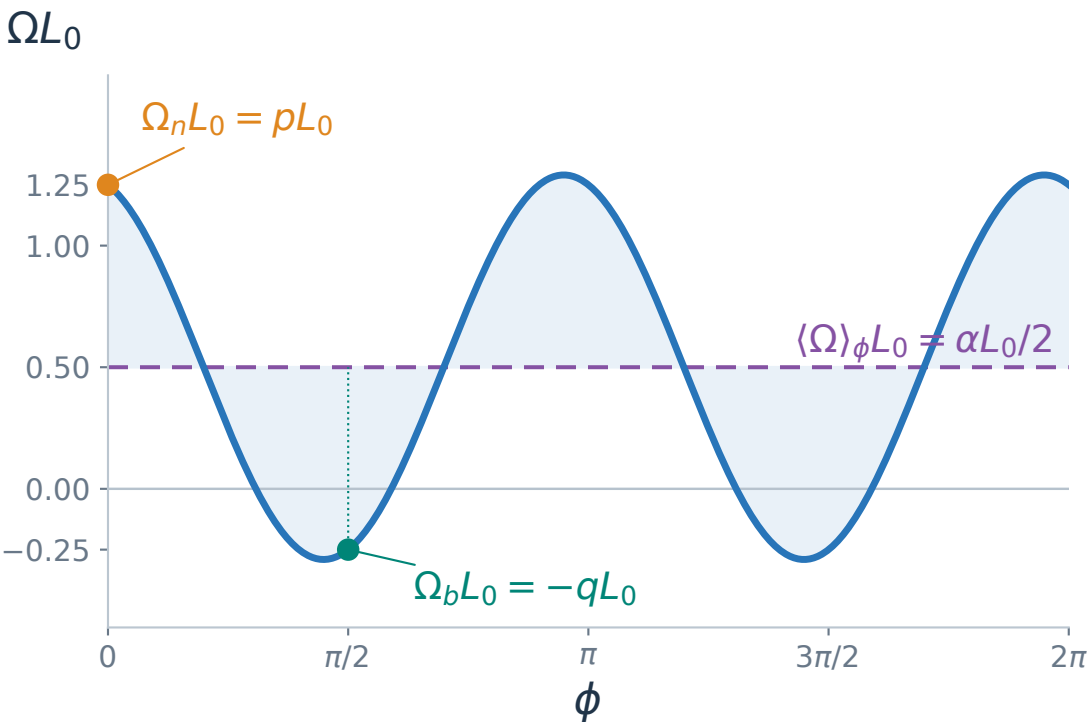


$$\Omega_n = \Omega(0), \quad \Omega_b = \Omega(\pi/2)$$

T out of page; positive: n to b

Circle: directions at s_0 , not a finite mapped boundary

3 UNIFORM AZIMUTHAL MEAN



A positive mean can include negative directions.

Shading: directional departure from the mean

L_0 : chosen reference length; ΩL_0 : dimensionless

SUM / FAC

$$\Omega_n + \Omega_b = \alpha = \frac{\mu_0 j_{\parallel}}{B}$$

$p + (-q)$

DIFFERENCE / SIGNED SHEAR

$$\Omega_n - \Omega_b = \beta_g = \frac{D}{B}$$

$p - (-q)$

MEAN / HALF OF ALPHA

$$\langle \Omega \rangle_{\phi} = \frac{1}{2\pi} \int_0^{2\pi} \Omega(\phi) d\phi = \frac{\alpha}{2} = \frac{\mu_0 j_{\parallel}}{2B}$$

Uniform directions at one point; not an along-line average

Illustration: $(\alpha, \beta_g, \delta_g) = (1, 1.5, 0.5)/L_0, \quad \theta = 0$

Arrows show instantaneous angular motion; radial deformation is omitted in panel 2.

Local thin-tube model; the planar reference line has zero torsion.

In a general Frenet frame:

$$d\phi/ds = \Omega(\phi) - \tau$$